

Machine Learning (CE 40717)

Fall 2024

Ali Sharifi-Zarchi

CE Department
Sharif University of Technology

November 7, 2024

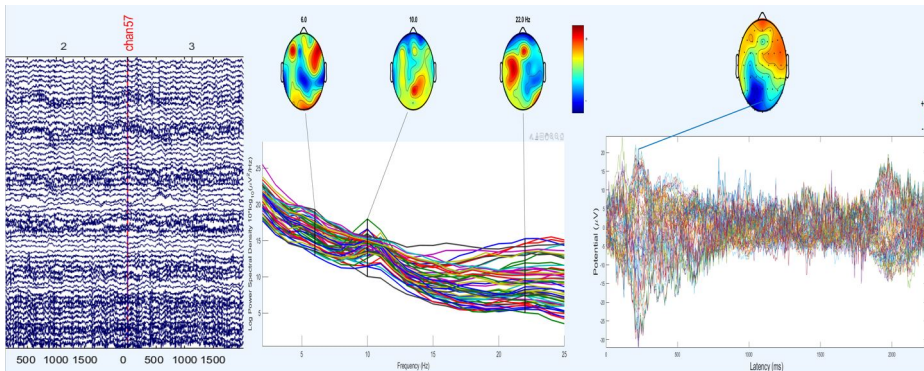


- 1 Introduction
- 2 Principal Component Analysis (PCA)
- 3 Choosing the Number of Principal Components
- 4 Applications
- 5 Shortcomings and Other Methods
- 6 Conclusion
- 7 References

7 References

High Dimensional Data

- High dimensions has many features.
- EEG signals from the brain, recorded with 56 channels and 3000 time points per trial.



High Dimensional Data

- Social media



Figure 1: Figure reference

100

- **Feature Selection**
 - Select a subset from a given feature set.
- **Feature Extraction**
 - A linear or non-linear transform from the original feature space to a lower dimension space.

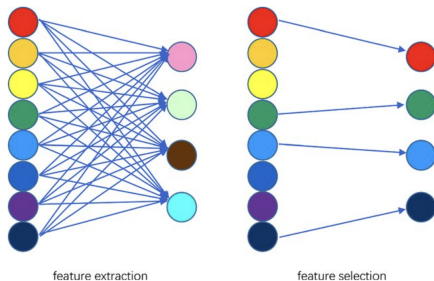


Figure 2: Figure reference

Dimensionality Reduction Purpose

- Maximize retention of **important information** while reducing dimensionality.
- What is **important information**?

Purpose: Variance of Data

- Maximize retention of **important information** while reducing dimensionality.
- **Information:** Variance of projected data

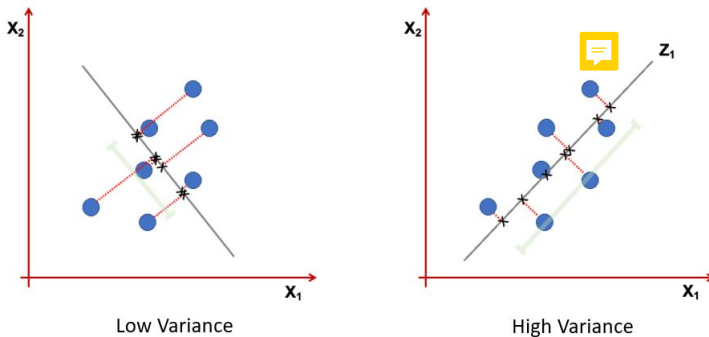


Figure 3: Figure reference

Purpose: Local Geometric Neighborhood

- **Information:** Preserve local geometric neighborhood.

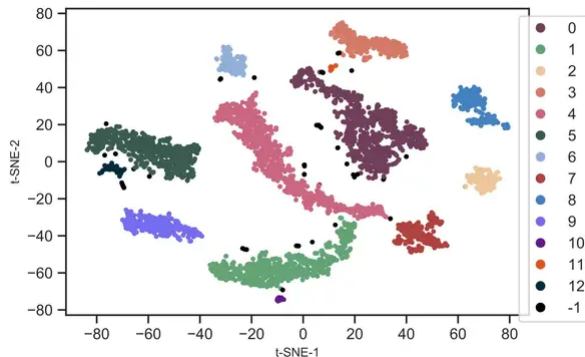


Figure 4: Figure reference

Purpose: Local and Global Geometric Neighborhood

- **Information:** Preserve both local and global geometric neighborhood.

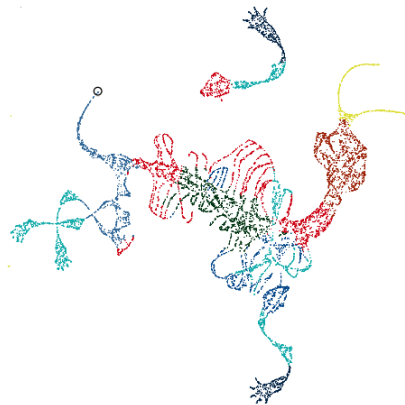


Figure 5: Figure reference

② Principal Component Analysis (PCA)

Background

Sample Covariance Matrix Algorithm

4 Applications

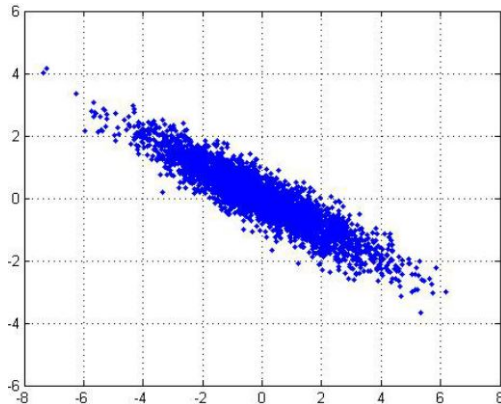
5 Shortcomings and Other Methods

6 Conclusion

7 References

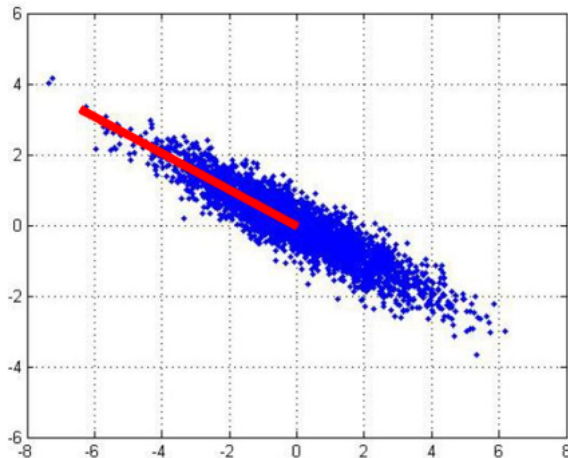
- Given data points in a d -dimensional space, project them into a lower dimensional space while preserving as much information as possible:
 - Find the best planar approximation of 3D data.
 - Find the best 12-D approximation of 104-D data.
- In particular, choose projection that minimizes the squared error in reconstructing the original data.

- 2D Gaussian dataset:



Principal Components Idea

- First PCA axis:

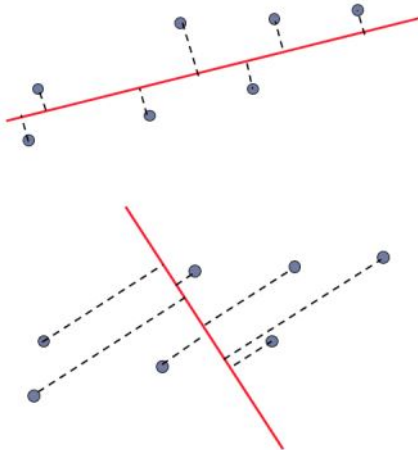


- First and second PCA axes:



Random vs. Principal Projection

- Random direction versus principal component:



Definition

- **Goal:** reducing the dimensionality of the data while preserving important aspects of the data.

- Suppose $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1^T \\ \vdots \\ \mathbf{X}_N^T \end{pmatrix}_{N \times d} = \begin{pmatrix} F_1 & F_2 & \dots & F_d \\ x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \vdots & \vdots & \vdots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{Nd} \end{pmatrix}$

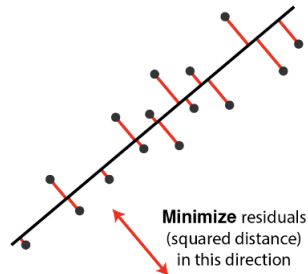
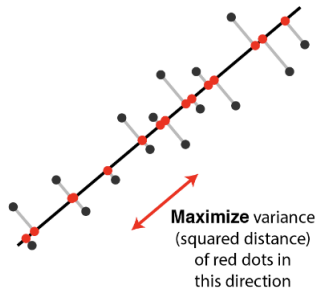
- $\mathbf{X}_{N \times d} \xrightarrow{\text{PCA}} \tilde{\mathbf{X}}_{N \times k}$ with $k \leq d$
- **Assumption:** Data is mean-centered, which is: $\mu_x = \frac{1}{N} \sum_{i=1}^N X_i = 0_{d \times 1}$



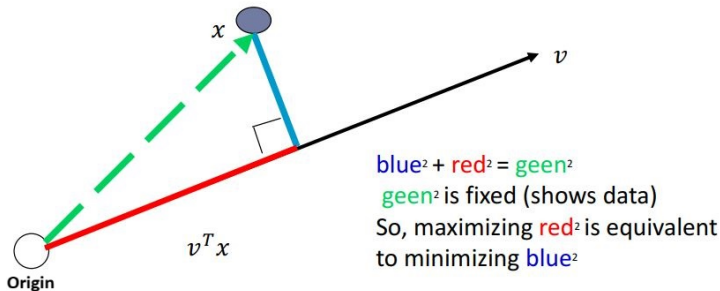
Interpretations

Orthogonal projection of the data onto a **lower-dimensional linear subspace** that:

- **Interpretation 1.** Maximizes variance of projected data.
- **Interpretation 2.** Minimizes the sum of squared distances to the subspace.



- 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99



Equivalence of the Interpretations

Principal Components (PCs): A set of **orthonormal** vectors ($v = [v_1, v_2, \dots, v_k]$) (where each v_i is $d \times 1$) generated by PCA, which fulfill both of the interpretations.

Interpretation 1. Maximizes variance of projected data

- Projection of data points on v_1

$$\Pi = \Pi_{v_1} \{X_1, \dots, X_N\} = \{v_1^T X_1, \dots, v_1^T X_N\}$$

- Note that $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$$\mathbb{E}[X] = 0 \rightarrow \text{Var}(\Pi) = \frac{1}{N} \left(\sum_{i=1}^N (v_1^T X_i)^2 \right)$$

2 Principal Component Analysis (PCA)

Sample Covariance Matrix Algorithm

4 Applications

5 Shortcomings and Other Methods

6 Conclusion

How to Find Eigenvalues and Eigenvectors?

- We know that

$$Av = \lambda v$$

- So

$$Av - \lambda v = 0$$

$$(A - \lambda I)v = 0$$

- v can not be zero, so:

$$\det(A - \lambda I) = 0$$

- Solve for λ
- Substitute λ back into the equation $Av = \lambda v$ to find v .

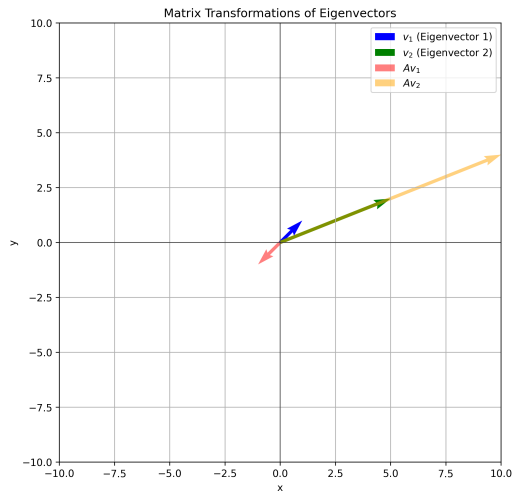
- Assume $A = \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$
- $A - \lambda I = ?$

Numerical Example (Continued)

- Assume $A = \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$
- $A - \lambda I = \begin{pmatrix} 4 - \lambda & -5 \\ 2 & -3 - \lambda \end{pmatrix}$
- Determinant $(A - \lambda I) = (4 - \lambda)(-3 - \lambda) + 10 = \lambda^2 - \lambda - 2$
- $\lambda = -1$ or $\lambda = 2$
- $\lambda_1 = -1$: $(A - \lambda_1 I)v_1 = \begin{bmatrix} 5 & -5 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- $\lambda_2 = 2$: $(A - \lambda_2 I)v_2 = \begin{bmatrix} 2 & -5 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow v_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

Visualization

- $Av = \lambda v$



What is Covariance?

- Covariance is a measure of how much two random features vary together.
- $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[(Y - \mathbb{E}[Y])(X - \mathbb{E}[X])] = \text{Cov}(Y, X)$
- So covariance is symmetric.
- Such as heights and weights of individuals.

What is a Covariance Matrix?

- A **covariance matrix** generalizes the concept of covariance to multiple features.
- For a random vector $\mathbf{F} = [F_1, F_2, \dots, F_d]$:

$$\Sigma = \begin{pmatrix} \text{Var}(F_1) & \text{Cov}(F_1, F_2) & \cdots & \text{Cov}(F_1, F_d) \\ \text{Cov}(F_2, F_1) & \text{Var}(F_2) & \cdots & \text{Cov}(F_2, F_d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(F_d, F_1) & \text{Cov}(F_d, F_2) & \cdots & \text{Var}(F_d) \end{pmatrix}$$

- The diagonal elements are the variances, and off-diagonal elements are covariances.

Covariance Matrix Example

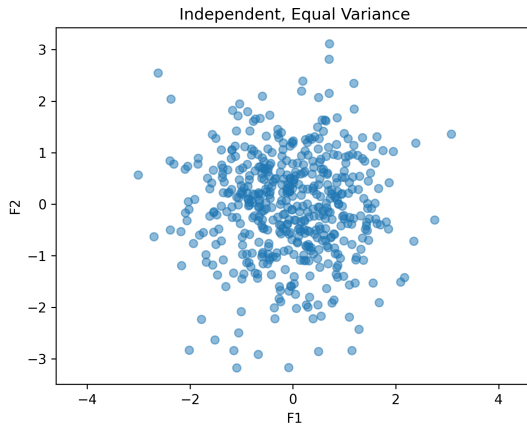
- Suppose there is two feature covariance matrix:

$$\Sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$$

- Why $b = c$?
- What is the relation between a , b , and d ?

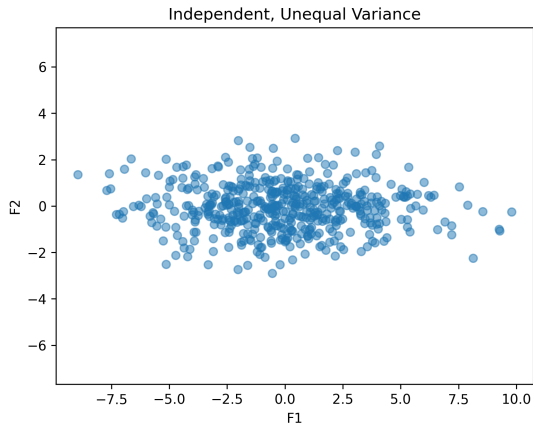
Covariance Matrix Example

- If $\Sigma = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$, then:



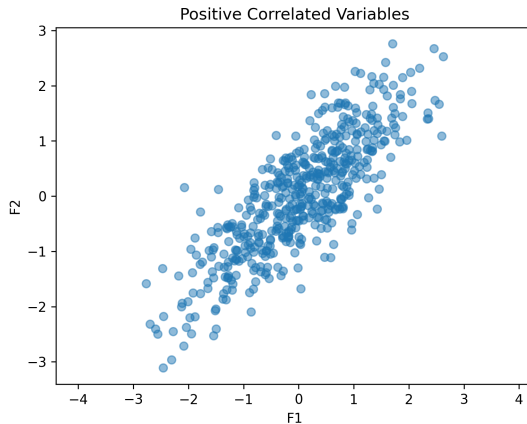
Covariance Matrix Example

- If $\Sigma = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$ and $a > d$, then:



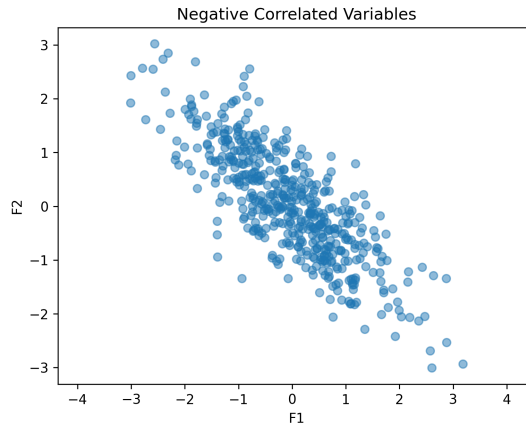
Covariance Matrix Example

- If $\Sigma = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, $a > d$, and $b > 0$, then:



Covariance Matrix Example

- If $\Sigma = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, $a > d$, and $b < 0$, then:



Sample Covariance Matrix

- In practice, we estimate covariance from sample data.
- **Sample Covariance Matrix:** Given N samples of d features, the sample covariance matrix Σ is:

$$\Sigma_{d \times d} = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X})^T$$

- Where X_i is the i -th sample, and $\bar{X}_{d \times 1}$ is the mean of the samples.

Example Calculation of Sample Covariance Matrix

- Suppose we have the following three samples each one having two features F_1 and F_2 :

Sample	F_1	F_2
X_1	3	3
X_2	4	7
X_3	5	8
$\bar{X}$	4	6

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X})^T = \frac{1}{2} \left(\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2.5 \\ 2.5 & 7 \end{pmatrix}$$

Lagrange Multiplier: Geometrical Interpretation

- We want to maximize $f(x)$ subject to $g(x) = 0$.
- The optimal point occurs where the gradient of $f(x)$ is proportional to the gradient of $g(x)$ (i.e., they are aligned or in opposite directions).

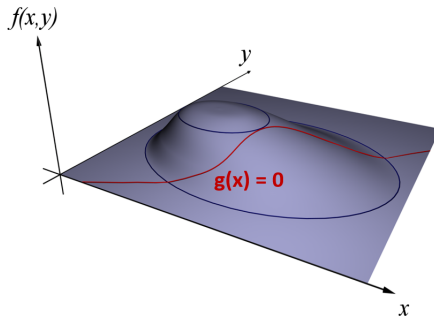


Figure 7: Figure reference

Lagrange Multiplier Method

- Combine the objective function and the constraint using the Lagrange function:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda g(x)$$

- Solve the system of equations:

$$\nabla \mathcal{L}(x, \lambda) = 0$$

- Where:
 - $\mathcal{L}$ is lagrangian
 - λ is lagrange multiplier
 - $g(x)$ is equality constraint
 - $f(x)$ is function

Example Problem

- Maximize:

$$f(x_1, x_2) = x_1 + x_2$$

- Subject to:

$$x_1^2 + x_2^2 = 1$$

- Lagrangian:

$$L(x_1, x_2, \lambda) = x_1 + x_2 - \lambda(x_1^2 + x_2^2 - 1)$$

- Partial derivatives:

$$\begin{cases} \frac{\partial L}{\partial x_1} = 1 - 2\lambda x_1 = 0 \\ \frac{\partial L}{\partial x_2} = 1 - 2\lambda x_2 = 0 \\ \frac{\partial L}{\partial \lambda} = -(x_1^2 + x_2^2 - 1) = 0 \end{cases}$$

Example Problem (Continued)

- Solving:

$$\begin{cases} \lambda x_1 = \frac{1}{2} \\ \lambda x_2 = \frac{1}{2} \\ x_1^2 + x_2^2 = 1 \end{cases}$$

- Since $\lambda x_1 = \lambda x_2$, then $x_1 = x_2$.
- Substitute $x_1 = x_2$ into the constraint:

$$2x_1^2 = 1 \implies x_1 = \pm \frac{1}{\sqrt{2}}$$

- Optimal solution:

$$x_1 = x_2 = \frac{1}{\sqrt{2}}, \quad f_{\max} = x_1 + x_2 = \sqrt{2}$$

Generalization to Multiple Constraints

- The Lagrange multiplier method can be extended to cases with more than one constraint.
- For constraints $g_1(x) = 0, g_2(x) = 0, \dots, g_m(x) = 0$, the Lagrangian becomes:

$$\mathcal{L}(x, \lambda_1, \lambda_2, \dots) = f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x) - \dots$$

1 Introduction

2 Principal Component Analysis (PCA)

Background

Sample Covariance Matrix Algorithm

3 Choosing the Number of Principal Components

4 Applications

5 Shortcomings and Other Methods

6 Conclusion

7 References

Step 1: Expression for Variance

- The variance of the projected data onto the direction v is:

$$\text{Var}(Xv) = \frac{1}{n} \sum_{i=1}^n (x_i^\top v)^2$$

- This can be rewritten as:

$$\text{Var}(Xv) = \frac{1}{n} \|Xv\|^2 = \frac{1}{n} v^\top X^\top X v = v^\top \Sigma v$$

Step 2: Maximization Problem

- We aim to maximize the variance $v^\top \Sigma v$ under the constraint that $\|v\| = 1$.
- This leads to the following optimization problem:

$$\max_v v^\top \Sigma v \quad \text{subject to} \quad \|v\| = 1$$

- 1000

Image Compression

- 



Image Compression

- 144D to 60D



Image Compression

- 144D to 16D



Image Compression

- The 16 most important eigenvectors

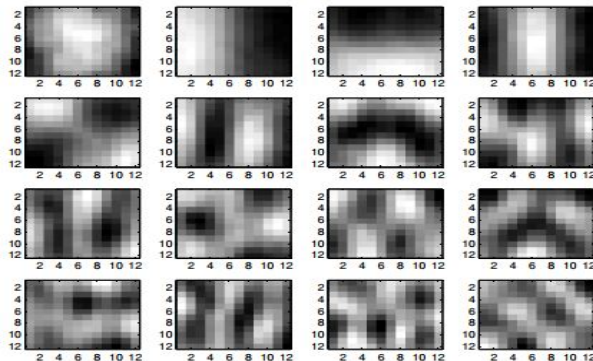


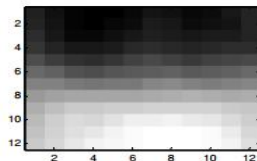
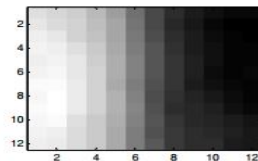
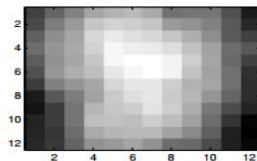
Image Compression

- 144D to 3D



Image Compression

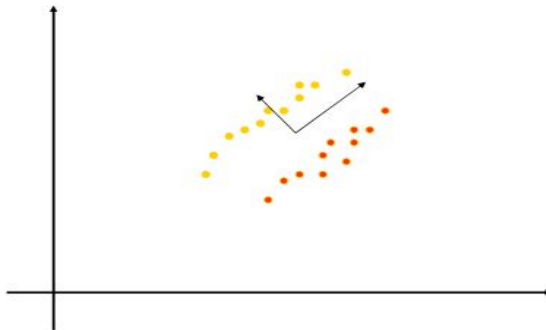
- The 3 most important eigenvectors



- 1 Introduction
- 2 Principal Component Analysis (PCA)
- 3 Choosing the Number of Principal Components
- 4 Applications
- 5 Shortcomings and Other Methods**
- 6 Conclusion
- 7 References

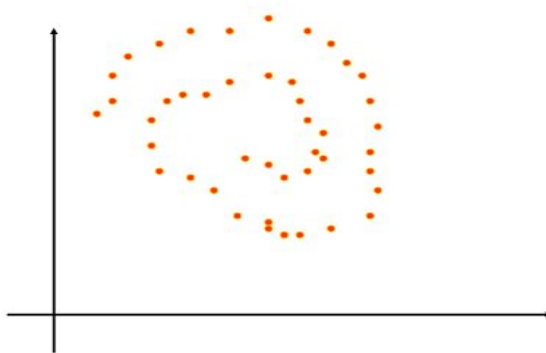
Class Labels

- PCA doesn't recognize about class labels.
- Alternative solution: Linear Discriminant Analysis (LDA)



Non-Linear

- PCA cannot capture non-Linear structure.
- Alternative solution: Kernel PCA



Conclusion

- PCA
 - Finds orthonormal basis for data
 - Sorts principal components in order of importance
 - Discards low significance principal components
- Applications
 - Get compact description
 - Remove noise
 - Improve classification (hopefully)
 - More efficient use of resources
 - Statistical
- Not magic
 - Doesn't recognize class labels
 - Can only capture linear variation

Contributions

- **This slide has been prepared thanks to:**
 - Mohammad Mowlavi

- [1] M. Soleymani Baghshah, “Machine learning.” Lecture slides.
- [2] B. Póczos, “Advanced introduction to machine learning.” Lecture slides.
CMU-10715.
- [3] M. Gormley, “Introduction to machine learning.” Lecture slides.
10-701.
- [4] M. Gormley, “Introduction to machine learning.” Lecture slides.
10-301/10-601.
- [5] F. Seyyedsalehi, “Machine learning and theory of machine learning.” Lecture slides.
CE-477/CS-828.
- [6] G. Strang, “Linear algebra and its applications,” 2000.